Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2025	ENGI3451-WE01

Title: (BEng/MEng)
Electronics and Communications 3

Time Allowed:	2 hours
Additional Material provided:	No
Materials permitted:	N/A
Calculators Permitted:	Those from the Casio fx-83 and fx-85 series.

Instructions to Candidates:	
	This is an in-person, invigilated exam. Answer ALL questions. Use a separate answer book for each question. All relevant workings must be shown.

Revision:	

Q1

(a) Consider double sideband amplitude modulation (DSB AM) with a carrier signal $c(t) = 3\cos(2.8 \times 10^6\ \pi\, t + 1.5)$ and a message signal $m(t) = 2\cos(3000\pi t) + 1.2\cos(6000\pi t)$. The resulting DSB AM signal is denoted by $s(t)$.

i. Provide the expression of $s(t)$. **[10%]**

ii. What is the bandwidth of $s(t)$? **[10%]**

iii. Explain whether we can recover $m(t)$ from $s(t)$ using an envolope detector. **[10%]**

iv. Explain how the power and bandwidth efficiencies of DSB AM could be improved. **[10%]**

(b) We aim to transmit three baseband (low frequency) signals, $m_1(t)$, $m_2(t)$ and $m_3(t)$, without interference using the frequency division multiplexing (FDM) technique. The frequency band of each baseband signal is 5 kHz. Single sideband amplitude modulation (SSB AM) with lower sideband is used. The available frequency band for transmission is 600 kHz–700 kHz.

i. Determine the smallest carrier frequencies that could be used for the three baseband signals. **[10%]**

ii. Provide the block diagram of the corresponding FDM transmitter. **[25%]**

(c) Consider the frequency demodulator shown in Figure Q1. The input signal to the frequency demodulator is: $s(t) = \cos\left(2 \times 10^8\ \pi\, t + 8\sin(6000\ \pi\, t)\right)$.

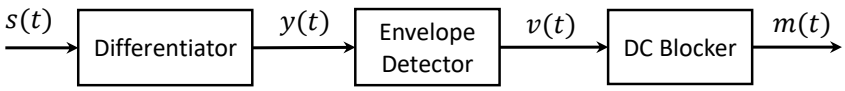


Figure Q1: A frequency demodulator.

i. Explain how the frequency modulator in Figure Q1 works. **[10%]**

ii. Provide the expression of $m(t)$ at the frequency demodulator's output. **[15%]**

Q2

(a) Consider a source that produces outputs a_1, a_2, a_3, a_4, a_5, a_6 and a_7 with probabilities $\frac{1}{3}$, $\frac{1}{3}$, $\frac{1}{9}$, $\frac{1}{9}$, $\frac{1}{27}$, $\frac{1}{27}$ and $\frac{1}{27}$, respectively. Assume coding over an alphabet of size $D = 3$ (ternary), for example, $\{0, 1, 2\}$.

i. What is the entropy of this source? **[10%]**

ii. Explain whether we can design a prefix code for this source with the following lengths for the codewords: $l_1 = 1$, $l_2 = 1$, $l_3 = 2$, $l_4 = 2$, $l_5 = 2$, $l_6 = 3$, $l_7 = 3$. **[10%]**

iii. Determine the average codeword length and the lengths of the codewords for both Shannon and Huffman codes without designing the codes. **[15%]**

(b) Consider a communication system that requires a (k, n) code with an error correction capability of at least 3 errors. Assume that a Hadamard code is used.

i. Determine the smallest n that satisfies this requirement. **[20%]**

ii. What is the largest value of k for this code? **[10%]**

iii. What is the rate of the designed code? **[5%]**

(c) A binary symmetric channel with a flipping probability of α is shown in Figure Q2.

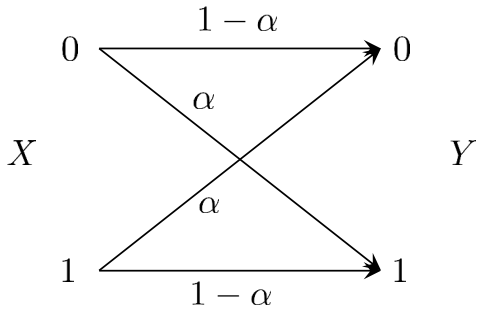


Figure Q2: A binary symmetric channel.

Determine the mutual information between X and Y, that is $I(X;Y)$, for $p_X(0) = 0.3$, $p_X(1) = 0.7$ and $\alpha = 0.1$. Note that $P_X(A)$ denotes the probability of event $X = A$. **[30%]**

Q3

(a) You are designing an amplifier chip. The circuit consists of three sub circuits; 1) A p-type MOSFET coupled to a resistor that is used as a current mirror; 2) A differential amplifier, that has two inputs and one output, this stage uses n-type MOSFETs; 3) Finally there is a common source amplifier which again uses an n-type MOSFET, this acts as a secondary gain stage. There is only one resistor used in the circuit. Draw this circuit diagram. Label each stage of the circuit. **[20%]**

(b) You decide you want to better understand the differential amplifier section of the circuit. Therefore you enter the circuit diagram for only the differential amplifier section of the circuit into LTspice. The transistors are considered perfect. Through experimentation you discover that the differential input voltage is related to the output current through the expression $I_{out} = g_m V_D$. Using diagrams with arrows showing the current flow explain why this is the case. **[20%]**

(c) Each of the transistors you choose for your differential amplifier has an effective channel resistance of 2 Ohms, and a trans-conductance of 5 mS. Calculate the output impedance of the circuit and therefore calculate the differential output voltage for an input voltage of 1 mV. **[20%]**

(d) Draw the diagram of a CMOS inverter coupled to a load capacitor and using a diagram explain why the power dissipation is $P = C_L V_{DD}^2 f_{CLK}$. **[20%]**

(e) Draw the circuit diagram for a NOR gate that uses a pull up/pull down network. How do pull up/pull down networks reduce power consumption? The output of the circuit is coupled to a load capacitor C. If each transistor has a resistance of R when turned on, state all the combinations of possible charge and discharge times the circuit can generate. **[20%]**

continued

Q4

(a) You are designing a circuit to receive a series of bits that have been transmitted over a long copper cable. You are worried about radio frequency (RF) interference, to solve this you decide to use a Schmitt trigger circuit. Draw the diagram of a Schmitt trigger along with the labeled hysteresis loop. Calculate the positive and negative threshold voltages, if only 1 kOhm resistors are used in the circuit and the comparator is run on a +5V/-5V supply. **[20%]**

(b) You now decided to develop a relaxation oscillator. The oscillator comprises of a comparator, a delay network that consists of a capacitor (C) and a resistor (R), and a feedback loop that has the resistors R_1 and R_F.

i. On the same graph, draw a waveform of the output voltage of this oscillator and also the waveform of the voltage across the capacitor. The relative magnitudes of each waveform should be considered carefully and maximum and minimum voltages labelled. **[20%]**

ii. Derive the equation relating the components of the circuit to the time constant of the oscillator. **[20%]**

(c) Describe using diagrams how an XOR gate can be used to provide feedback to a phase detection circuit. **[20%]**

(d) Draw the diagram for a cross coupled LC oscillator circuit that is actively driven by MOSFETs. The circuit should be constructed in such a way that the output voltage can not go above V_{DD}. **[20%]**

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